Juan de Dios Alvarez

Data Scientist / IA

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Profile

Data Scientist with over three years of continuous, hands-on delivery of Machine Learning, Deep Learning, and Generative AI systems, embedded in a 20+ year technical trajectory spanning research computing, IT infrastructure, and consulting. Recent work centers on production-oriented ML: transformer-based NLP and RAG pipelines, satellite-image-based crop yield modeling deployed into a governed data infrastructure, and independent exploratory work in document AI, embeddings, and unsupervised clustering. Comfortable across the full model lifecycle -- data preparation, model development, productive deployment, and stakeholder-facing delivery for non-technical audiences. Background in academic research (HAWC Collaboration, CERN) gives the analytical rigor; the last three years have been about turning that rigor into deployed systems that a business actually runs on.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

May 2026 -- present

Self-directed technical practice following the FIRA engagement, split between a multi-repo business-systems build for a small operating business (most active and current stream) and exploratory ML/NLP prototyping. Designed and continue to actively develop a multi-repository system automating supplier/tax reconciliation and product cataloging, with standardized data contracts between components. In parallel, built and iterated a document OCR/extraction comparison harness (classical, LLM-based, and layout-aware strategies) against real scanned ledger data; earlier-stage exploratory work also includes a local PII-anonymization design and a sentence-embedding/clustering approach for unsupervised text categorization, both at proof-of-concept stage.

- Designed and actively maintain a multi-repository business-automation system: supplier/tax reconciliation and product catalog generation, each with a standardized output contract into a shared warehouse
- Built and iterated an OCR/document-AI comparison harness (classical engines, LLM-based extraction, layout-aware segmentation) against real scanned financial documents, with a working review tool for human curation
- Designed (proof-of-concept stage) a local, reversible PII-anonymization approach for Mexican-Spanish legal text (LFPDPPP / ISO 29101-aligned) and a sentence-embedding/clustering approach for unsupervised text categorization

Data Scientist | FIRA

January 2025 -- January 2026

Led a digital transformation initiative for FIRA's Data Governance team: replaced a manual, Excel-based analytical process with an automated, governed data infrastructure supporting agricultural credit assignment and non-payment risk assessment. Built 74 extraction scripts across 9+ public and institutional data sources, loading into PostgreSQL (JSONB, materialized views, 40+ custom functions) on a scheduled, unattended pipeline, with 15+ dashboards served through Apache Superset. Applied a Random Forest model combining two satellite sources (Sentinel-2, Planet Labs) to estimate corn production from multispectral imagery across 310 georeferenced field polygons (R2 = 0.928). Led an undergraduate student team contributing to the pipeline under supervision.

- Migrated a single-analyst, Excel-based process to a governed, reproducible pipeline any team member could run and extend
- Built and scheduled 74 extraction scripts (Python, Playwright, aiohttp) across 9+ heterogeneous sources into a PostgreSQL/PostGIS warehouse
- Applied a satellite-imagery ML model (Random Forest, Sentinel-2 + Planet Labs, vegetation indices) to estimate crop yield at the field level
- Delivered 15+ dashboards via Apache Superset, replacing a single-specialist bottleneck with team-wide access

- Defined governance standards: version control (GitLab), extraction metadata, reproducibility conventions
- Led a student team ("residencias profesionales") as part of the implementation effort

Data Scientist | CITEIM

October 2024 -- January 2025

Led the technical foundation for a Mexican Sign Language (LSM) recognition and translation system, a domain with limited prior Spanish-language research. Conducted a structured state-of-the-art review to identify a defensible technical direction, then designed a two-stage transformer-based pipeline (gloss detection, then translation to written Spanish) architected for fully local, on-device deployment -- evaluating LLM integration under that local-only constraint to reduce latency and avoid cloud dependency. Built dataset tooling (automated search/download, normalization, keypoint-based quality verification) to address initial data scarcity, producing a controlled video dataset for model development.

- Directed a state-of-the-art and competitive review that set the technical direction before implementation
- Designed a two-stage transformer-based pipeline for sign language recognition and translation, architected for local, cloud-free deployment
- Built automated dataset expansion, normalization, and quality-verification tooling to overcome data scarcity in an under-resourced language domain

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

Led data analysis for business KPI reporting and quality dashboards (R, Python, Tableau, Looker). Prototyped real-time data processing from WebSocket streams for financial market visualization on AWS. Founded and led a student Data Science group applying analytical skills to real problems for nonprofit organizations.

- Built KPI and quality dashboards consumed directly by business stakeholders
- Prototyped streaming data pipelines for financial market analysis on cloud infrastructure
- Founded a mentorship program pairing skill development with social-impact data work

Data Scientist | BairesDev

September 2023 -- November 2023

Delivered a Retrieval-Augmented Generation (RAG) system for a legal services client: transformer-based NLP over unstructured contract text, enabling each lawyer to query their own portfolio and extract obligations, risk clauses, and precedents without manual review. Also applied Neo4j graph modeling to reveal non-obvious relationships in complex datasets and built transformer-based text summarization models.

- Designed and delivered a RAG pipeline for legal contract analysis: classification, obligation extraction, risk flagging
- Applied graph database modeling (Neo4j) and transformer-based summarization as parallel technical deliverables
- Demonstrated that a well-scoped, six-week engagement can deliver a high-value Generative AI proof of concept

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

Joined as the company's first data scientist to build its analytics practice from the ground up, expanding beyond Master Data Management into data insights and analytics for international (US-based) clients. Delivered end-to-end solutions: data lake construction, automated ingestion, ETL, NLP-based classification (Azure ML), forecasting, and executive dashboards. Built a public-records NLP pipeline for a client (school board meeting minutes) that went from proof of concept to full production deployment, feeding a Power BI dashboard used directly in sales strategy. Earned Semarchy xDM and AWS Cloud Practitioner certifications during this period.

- Took an NLP/ML classification pipeline from proof of concept to production deployment, generating client-facing competitive intelligence from public data
- Applied NLP and ML classification to unstructured text using Azure Machine Learning
- Implemented ETL and Master Data Management (Semarchy xDM, Tibco EBX) on AWS, including data profiling, lineage, and Golden Record production

- Built forecasting models and executive dashboards (Power BI, Tableau) directly informing client strategy

Education

PhD Candidate in Physics University of Michoacan, Mexico | 2009 -- 2013 Focus: distributed computing and large-scale data analysis (HAWC Collaboration); Python and C++ analysis pipelines. Co-author on 14 publications.

Master's in Computer Science National Institute of Astrophysics, Mexico | 2006 -- 2009 Thesis: automatic text classification using prototype-based class reduction. Focus: Machine Learning, NLP.

Bachelor's in Physics and Mathematics University of Michoacan, Mexico | 1998 -- 2003

Certifications

- **Semarchy xDM** -- Master Data Management: Data Profiling, Lineage, Quality, Match/Merge, Golden Records, Governance
- **AWS Cloud Practitioner**
- **AWS Fundamentos de IA Generativa**

Technical Skills

Languages: Python, R, SQL, Java, C/C++ **ML / DL / GenAI:** Transformers, LLMs, RAG pipelines, scikit-learn, pandas, NumPy, sentence embeddings, unsupervised clustering (HDBSCAN, FINCH) **NLP:** Text classification, summarization, information retrieval, contract/document analysis **Computer Vision / Geospatial:** Satellite imagery modeling (Sentinel-2, Planet Labs), vegetation indices, QGIS, PostGIS **Data Management:** ETL, Master Data Management (Semarchy xDM, Tibco EBX), data lakes, data governance **Databases:** PostgreSQL, Oracle, MySQL, Neo4j **Cloud:** AWS (EC2, SageMaker), Azure Machine Learning **Visualization:** Apache Superset, Power BI, Tableau, Looker **Methodologies:** Agile (Scrum), MLOps practices, data governance, team leadership and mentoring

Languages

- **Spanish:** Native
- **English:** Professional proficiency
- **French:** Basic